

Developmental plasticity in thermal tolerance

Ontogenetic variation, persistence, and future directions



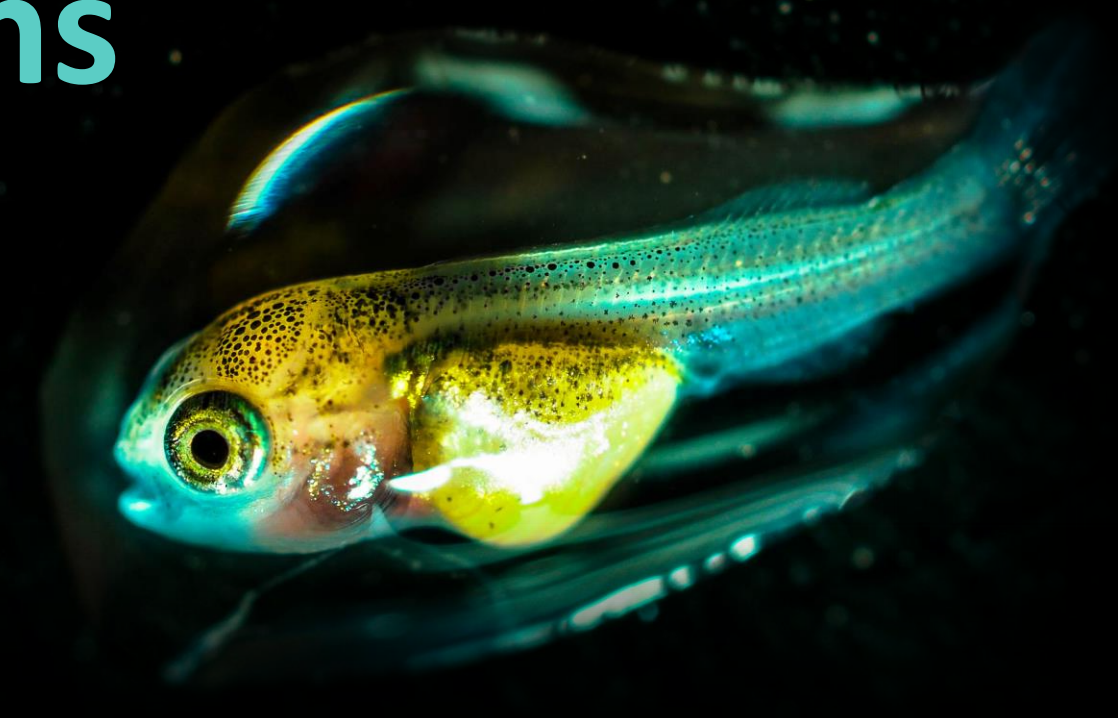
Patrice Pottier, Samantha Burke, Rose Y. Zhang, Daniel W.A. Noble, Lisa E. Schwanz, Szymon M. Drobniak, & Shinichi Nakagawa

@Patrice_Science

p.pottier@unsw.edu.au

0000-0003-2106-6597

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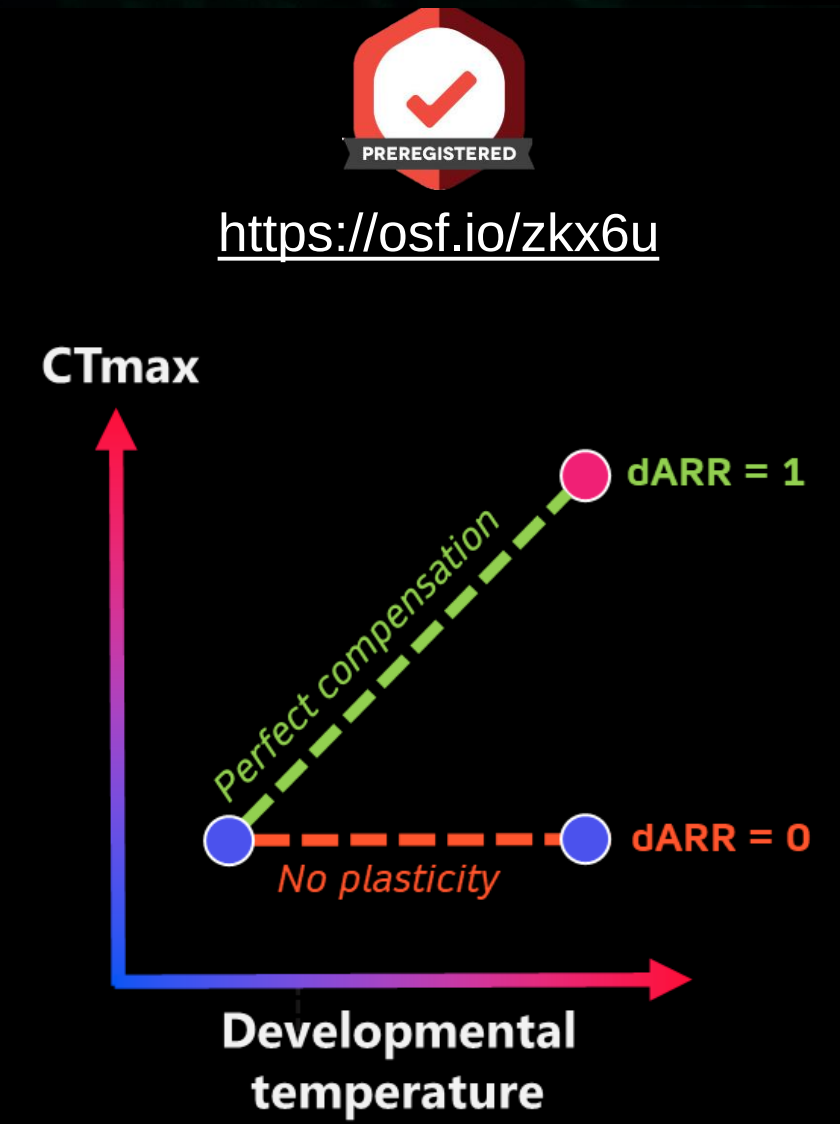
BACKGROUND

- Ectotherms can adjust their **thermal tolerance** by **acclimating** to new temperatures (plasticity)
- Temperatures** experienced during **development** can profoundly influence later phenotypes
- However, the role **embryonic and juvenile developmental plasticity** plays in allowing populations to cope with thermal extremes is poorly understood
- AIM 1:** Map the **current state-of-knowledge**
- AIM 2:** **Meta-analyse** how thermal tolerance is **initially** and **persistently** impacted by early thermal environments

eseb

METHODS

- Systematic review
 - Ectothermic animal
 - Thermal tolerance using **CTmax** or **LT50**
 - ≥ 2 developmental** temperatures
- 1089 effect sizes** from
 - 150 experimental studies**
 - 138 ectothermic species**
- Effect size = developmental acclimation response ratio (**dARR**)



RESULTS

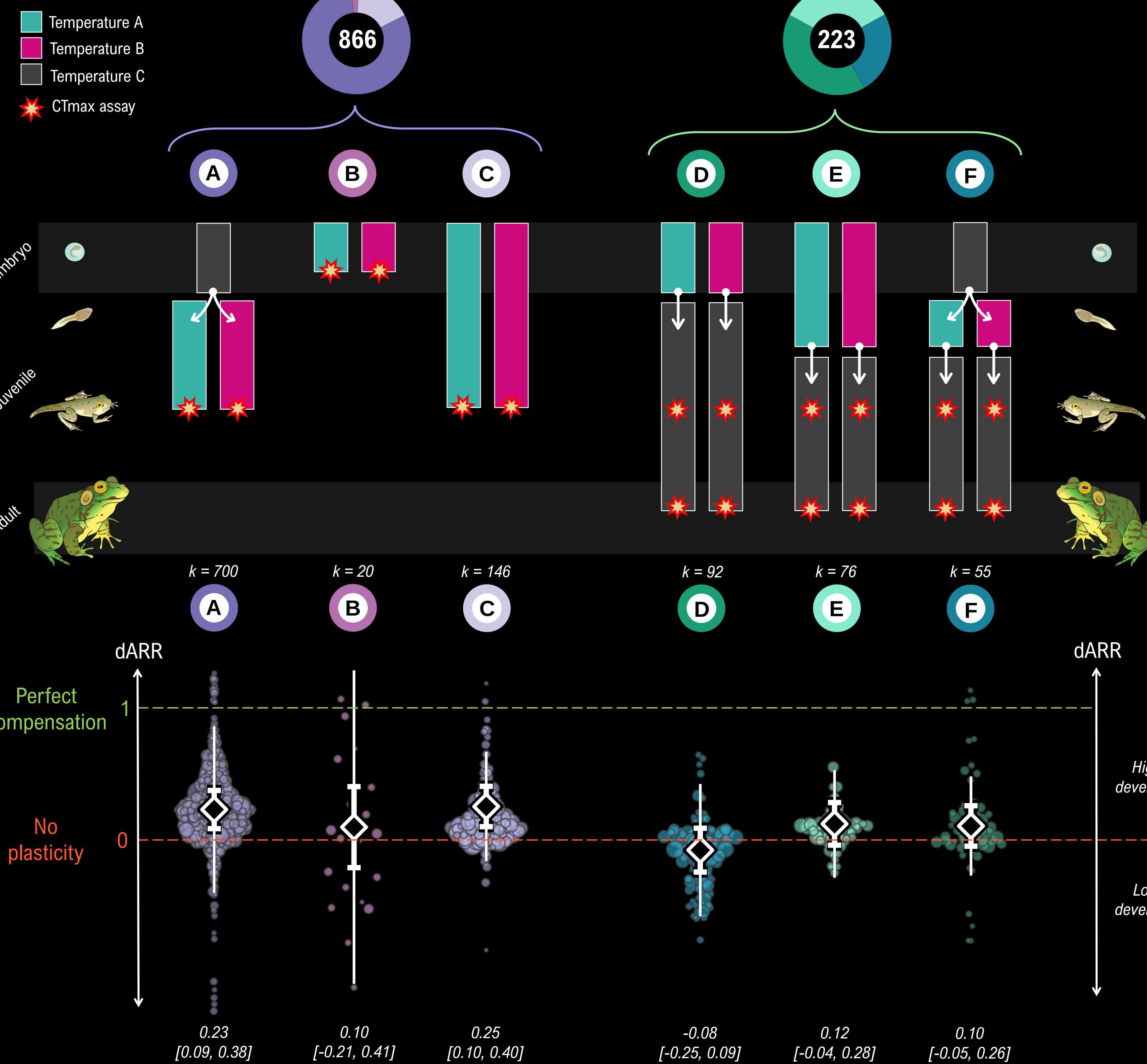
Ontogenetic variation & persistence

Without re-acclimation to a common garden condition

With re-acclimation to a common garden condition

Initial effects

Persistent effects



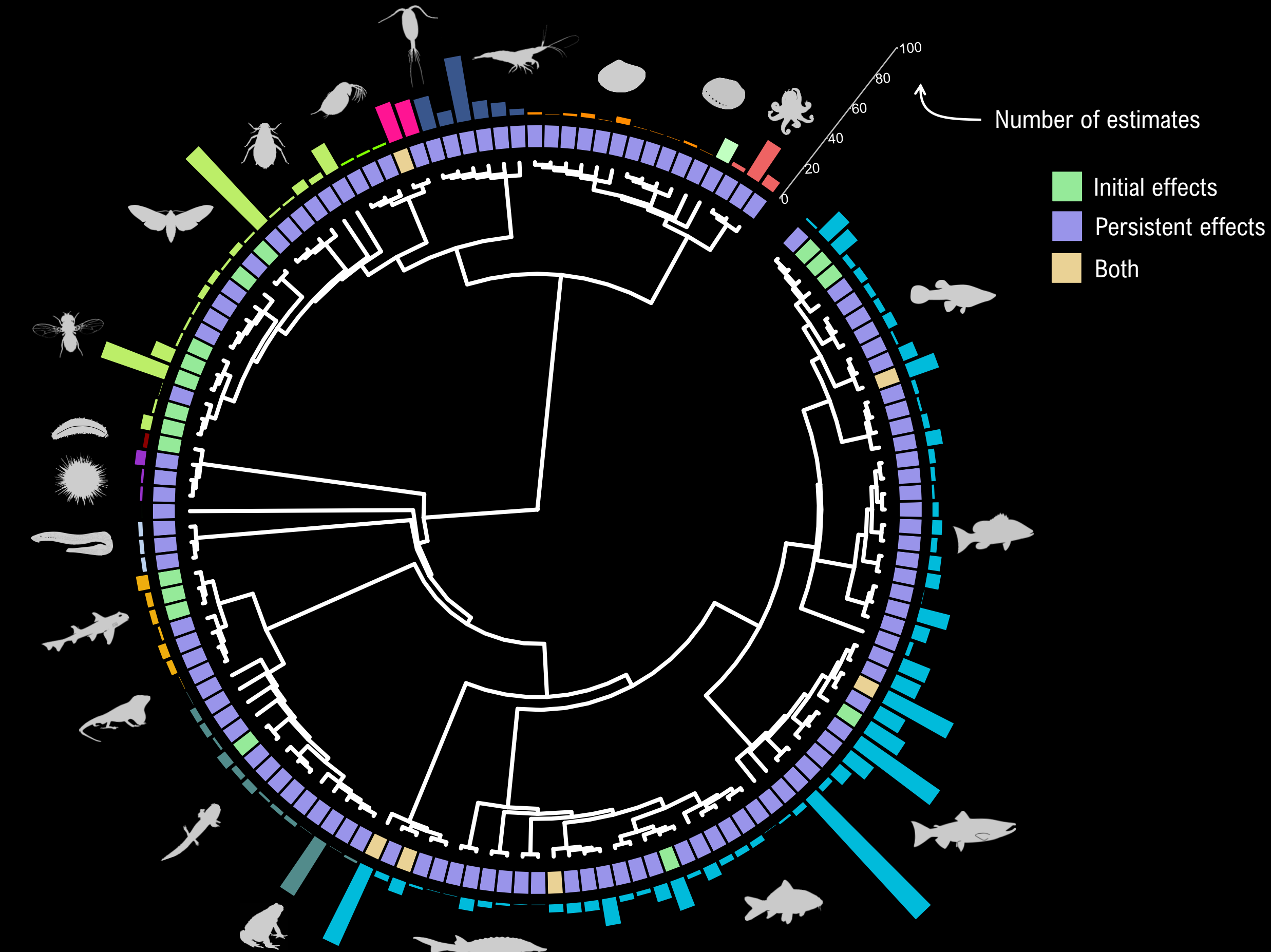
Weak overall responses for all developmental stages (A-C)

Embryos express **weaker** and **more variable** responses than juveniles (B)

Higher **embryonic** temperatures tend to **persistently reduce** the thermal tolerance of later life stages (D)

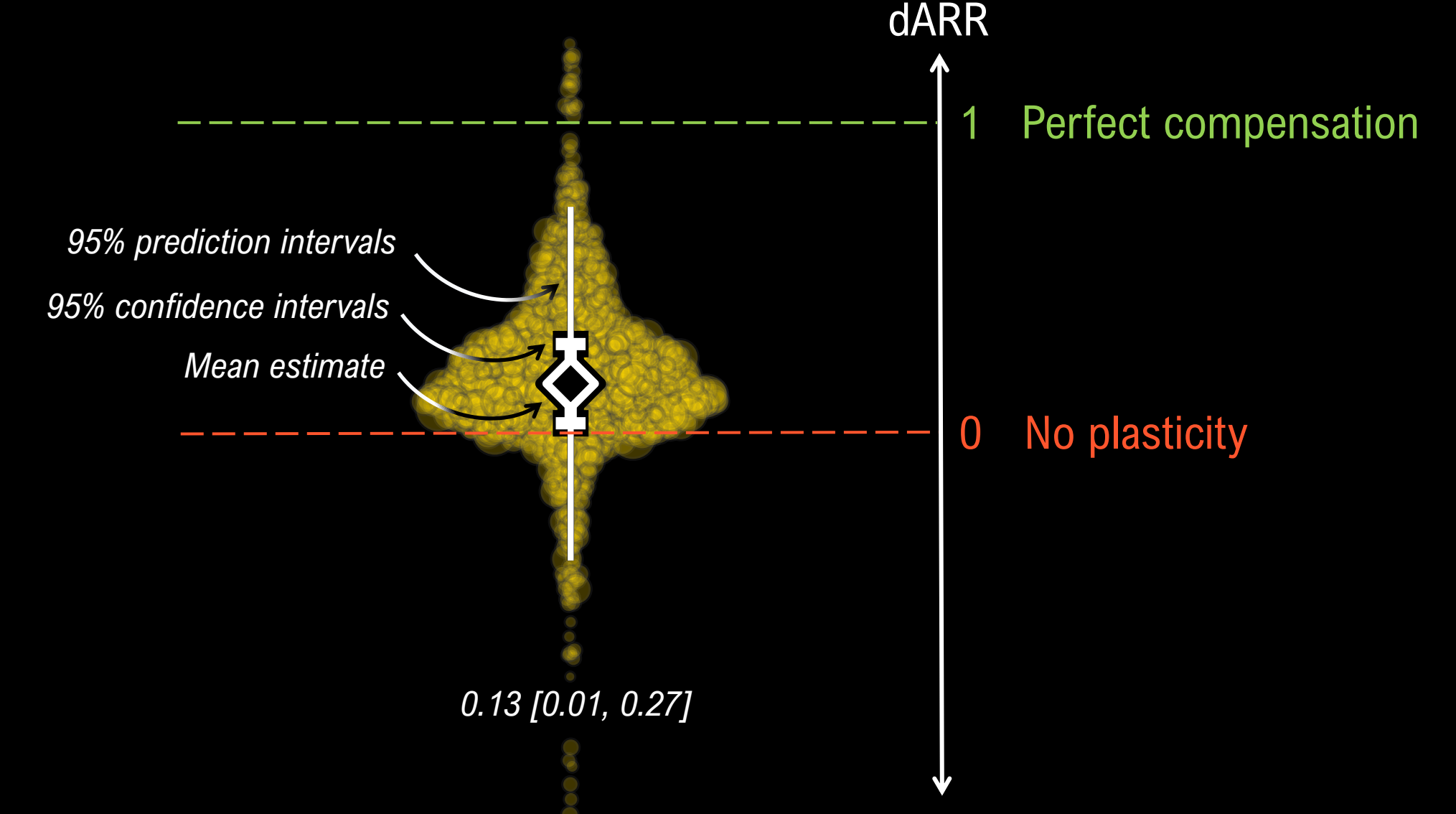
Higher **juvenile** temperatures tend to **persistently increase** the thermal tolerance of later life stages (E-F)

Taxonomic coverage



Persistent effects rarely tested across the tree of life
Dataset dominated by **fish** species (57% of the estimates)

Overall effects



High heterogeneity ($I^2 = 99\%$)

Weak responses to developmental temperatures overall

To increase heat tolerance by 1°C, it requires a **change** in temperature of 7.5 °C

KEY FINDINGS & PERSPECTIVES

Very **high heterogeneity**

More research is needed on **invertebrates** and the **herpetofauna**

Persistent effects are vastly **under-studied**

The **magnitude** and **direction** of persistent effects **varied with ontogeny**

Embryonic stages may represent a **critical window of vulnerability** to changing environments

Developmental changes in thermal tolerance **rarely reach levels of perfect compensation** and are **not sufficient** to buffer ectotherms from rising temperatures

Acknowledgements



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